

Thin double layer theory of the wide-frequency range dielectric dispersion of suspensions of non-conducting spherical particles including surface conductivity of the stagnant layer

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Abstract

The well-known results of the Dukhin–Shilov theory of the thin diffuse double layer polarization in low-frequency alternating fields is generalized in two directions: to include the surface conductivity of the stagnant layer and to describe a wide frequency range including both the low-frequency and the Maxwell–Wagner–O’Konski dispersions. The generalized expression obtained for the induced dipole moment is confirmed by comparison with numerical results of the DeLacey and White approach. Expressions describing the wide-range spectra of dielectric permittivity and conductivity of suspensions are also obtained. The results are presented in the terms of the ζ -potential and the parameter κa , taking also into account the surface conductivity behind the plane of shear. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

The investigation of the dielectric dispersion of suspensions involves determinations of its dielectric constant $\varepsilon(\omega)$ and conductivity $K(\omega)$ as functions of the angular frequency ω of the applied AC field. These quantities are related to the properties of the constituents of the suspension: the

bulk, surface, and geometrical characteristics of the disperse particles, their concentration (usually expressed in terms of volume fraction ϕ_p), and the characteristics of the dispersion medium (electrolyte solution viscosity, ion concentrations, valences and diffusion coefficients of all the ions).

The dielectric constant is the macroscopic manifestation of the electrical polarization of the suspension components. Different times are necessary for the development of the various mechanisms contributing to the system’s polarization. Therefore, the investigation of the frequency dependence of the dielectric permittivity and con-

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ductivity makes it possible to obtain information about the characteristics of the disperse system that are responsible for the polarization of the particles.

For suspensions in electrolyte solutions there are two characteristic mechanisms of dielectric dispersion, which are both sensitive to the polarization of the electric double layer of the disperse particles. The first is the low-frequency dielectric dispersion, caused by the influence of field-induced electrolyte concentration variations on the local electric currents in the double layer (concentration polarization of the double layer, or volume diffusion mechanism of double layer polarization). The second is the well-known Maxwell-Wagner dielectric dispersion, caused by the formation of the field-induced free (ionic) charge distributions near the surfaces that separate the phases with different local conductivities and dielectric constants.

The theory of the low-frequency dielectric dispersion has been first elaborated in [1–3] for dilute suspensions of homogeneous spherical particles with no volume conductivity and surrounded by a thin double layer. It is based on the well-known system of electrokinetic equations [4] where it is considered that the liquid and ions are immobile in the dense part of the double layer while, in its diffuse part, the viscosity, diffusion coefficients, and dielectric constant, are considered to have the same values as in the bulk electrolyte solution (the so-called *standard model* of the electric double layer). The theory was later confirmed [5] and generalized in different directions: for rotating applied fields [6], for particles with volume conductivity [7–9], for non-spherical particles [10–13], for concentrated suspensions [14–16], for weak electrolytes [17,18].

In the original Maxwell–Wagner theory [19,20], the surface was considered only as a geometrical boundary between homogeneous phases without any specific properties. This was generalized by O’Konski [21], who completed the model by taking into account the surface conductivity K^s , without any specification of its physical nature. Assuming that the surface conductivity is due to excess ion concentration in the proximity of the charged particle surface, it opened the possibility

for the characterization of the double layer using measured parameters of the Maxwell–Wagner (or rather Maxwell–Wagner–O’Konski) dielectric dispersion.

All analytical theories of the dielectric dispersion of suspensions deal separately with the MW dispersion or the low-frequency one. No formula is available that considers the wide frequency interval including both mechanisms, except for those presented by O’Brien [22] and by Grosse [23], which are based on simplified treatments of the low-frequency dispersion. The present work is devoted to the deduction of such an analytical formula that includes, furthermore, the contribution of the surface conductivity behind the slipping or shear plane (the so-called *generalized standard model* [24]).

A very important progress in the theory of both the low-frequency and the MW dielectric dispersions is due to the numerical theory by DeLacey and White [25], which was developed in the frame of the standard system of electrokinetic equations for dilute suspensions of non-conducting spherical particles in the absence of surface conductivity behind the slipping plane (*standard model*), and without any restriction on the diffuse layer thickness or the ζ -potential values. These numerical results will be used in this work to check the range of applicability of the analytical theory.

2. Theory

2.1. The natural frequency scale for suspensions in electrolyte solutions

The frequency range in which the dielectric dispersion of suspensions in electrolyte solution is usually measured roughly extends between 1 kHz and several hundred MHz. In order to define in this frame the *low frequency* and *high-frequency* ranges, it is useful to introduce an important concept dealing with a point in the frequency scale. This point corresponds to the *reciprocal of the relaxation time of the electrolyte solution* τ_m :

$$\omega_m \equiv \frac{1}{\tau_m} \equiv \frac{K_m}{\epsilon_m} \quad (1)$$

where K_m and ε_m are the conductivity and absolute dielectric constant of the dispersion medium. The value of τ_m characterizes the time required for the screening of charge and electric field perturbations in the electrolyte solution. Its role in the time dimension is similar to that played by the Debye screening length

$$L_D \equiv 1/\kappa = \sqrt{\frac{(D^+ + D^-)\varepsilon_m}{2K_m}} \quad (2)$$

in the space dimension (here $D^\pm$ are the ion diffusion coefficients and a symmetric electrolyte is assumed). The order of magnitude of τ_m corresponds to the time of ion diffusion through a distance equal to the Debye length:

$$\tau_m \approx \frac{L_D^2}{2D} \equiv \frac{1}{2D\kappa^2} \quad (3)$$

where D is the average ion diffusion coefficient.

It is so advisable to define the low-frequency range by the inequality:

$$\omega < < \omega_m. \quad (4)$$

For these frequencies, the characteristic value of the *conduction current density* in the electrolyte solution exceeds the *displacement current density*, and the space distribution of the local electric fields in the disperse system is mainly determined by the distribution of ionic currents. In the high-frequency range determined by the inequality:

$$\omega > > \omega_m \quad (5)$$

the characteristic value of the displacement current density exceeds that of the conduction current, and the space distribution of the local electric fields is determined by the polarization of the molecular dipoles, rather than by the distribution of ionic currents.

In the forthcoming considerations it is convenient to represent the time dependence of the harmonic external field $E(t)$ with frequency ω , by means of the complex multiplier $e^{i\omega t}$

$$E(t) = E(\omega) \cdot e^{i\omega t} \quad (6)$$

The well-known advantage of this representation is that all the field-induced magnitudes $X(t)$ have also this same time dependence (in the linear approximation with respect to the external field):

$$X(t) = X^*(\omega) \cdot e^{i\omega t} \quad (7)$$

where the asterisk denotes a complex quantity. If any magnitude X had a relaxation at a frequency close to ω , it would be out of phase with respect to the external field. In the representation used, this corresponds to the appearance of an imaginary part of $X^*(\omega)$ and also to its frequency dependence.

Using the representation given in Eqs. (6) and (7), the full electric response of the disperse system can be characterized by its complex permittivity

$$\varepsilon^*(\omega) = \varepsilon(\omega) - \frac{iK(\omega)}{\omega} \quad (8)$$

or its complex conductivity $K^*(\omega)$

$$K^*(\omega) = K(\omega) + i\omega\varepsilon(\omega). \quad (9)$$

It should be emphasized that the suspension permittivity $\varepsilon(\omega)$ and conductivity $K(\omega)$ are real functions of the frequency.

The contributions $\delta\varepsilon^*(\omega)$ and $\delta K^*(\omega)$ of the dispersed particles to the complex permittivity and conductivity of a dilute suspension are defined as:

$$\delta\varepsilon^*(\omega) \equiv \delta\varepsilon(\omega) - \frac{i\delta K(\omega)}{\omega} = \varepsilon^*(\omega) - \varepsilon_m^*(\omega) \quad (10)$$

$$\delta K^*(\omega) \equiv \delta K(\omega) + i\omega\delta\varepsilon(\omega) = K^*(\omega) - K_m^*(\omega) \quad (11)$$

where the functions

$$\varepsilon_m^*(\omega) = \varepsilon_m - \frac{iK_m}{\omega} \quad (12)$$

$$K_m^*(\omega) = K_m + i\omega\varepsilon_m \quad (13)$$

represent the complex permittivity and complex conductivity of the dispersion medium (ε_m , K_m , $\delta\varepsilon(\omega)$, and $\delta K(\omega)$ are real).

The functions $\delta\varepsilon^*(\omega)$ and $\delta K^*(\omega)$ are determined by the superposition of the long-range electric fields of the polarized particles. This can be expressed through the dipole coefficient $\gamma_p^*(\omega)$ of a particle and the volume fraction ϕ_p :

$$\delta\varepsilon^*(\omega) = 3\phi_p\varepsilon^*(\omega)\gamma_p^*(\omega) \quad (14)$$

$$\delta K^*(\omega) = 3\phi_pK_m^*(\omega)\gamma_p^*(\omega) \quad (15)$$

The dipole coefficient $\gamma_p^*(\omega)$ appears in the expression for the electric potential distribution φ_m^* , valid in the electro-neutral electrolyte solution outside the double layer at large distances r from the center of the particle:

$$\varphi_m^* = -Er \cos \theta + \frac{\gamma_p^* a^3 E}{r^2} \cos \theta \quad (16)$$

where a is the particle radius. The first term in the righthand side of this expression is the potential of the homogeneous external field, and θ is the angle between the radius vector of a point and the direction of the field. The second term in Eq. (16) represents the long-range potential distortion produced by the polarized particle together with its double layer.

According to [1–3,5,25], the frequency dependencies of the suspension permittivity and conductivity $\varepsilon(\omega)$ and $K(\omega)$ (in the frequency range where the characteristics of the dispersion medium may be considered as frequency-independent) are caused by the frequency dependence of the dipole coefficient $\gamma_p^*(\omega)$. Note that in the low-frequency part of the natural scale where the strong inequality (4) holds, the absolute value of the imaginary part of the complex permittivity of the dispersion medium (K_m/ω), exceeds its real part by a large factor. This results in the well-known very large values of the low-frequency suspension's permittivity due to the contribution of $\frac{K_m}{\omega} \operatorname{Im} \gamma_p^*(\omega) = \varepsilon_m \frac{\omega_m}{\omega} \operatorname{Im} \gamma_p^*(\omega)$ in the real part of Eq. (14).

It is interesting to point out that a symmetrical phenomenon, consisting in a very large contribution of the particles to the suspension conductivity, $\delta K(\omega)$, may be expected in the high-frequency part of the natural scale ($\omega > \omega_m$). It is due to the contribution of the addend $\varepsilon_m \omega \operatorname{Im} \gamma_p^*(\omega)$ that can be very large in the case that there is some mechanism that causes a frequency dependence of $\gamma_p^*(\omega)$ in the high-frequency range.

2.2. Middle-frequency range:

Maxwell–Wagner–O’Konski relaxation

There are various mechanisms contributing to the particle polarization, and each of them is

always associated to some property that differs for the solid, the liquid, and their interface. As mentioned above, the most widely known mechanism of dielectric dispersion, the *Maxwell–Wagner dispersion*, occurs when the phases in contact have different volume conductivities and dielectric permittivities. Let K_p and ε_p denote the volume conductivity and permittivity of the particles. If the ratio between these two quantities differs for the particles and the medium, i.e. if

$$\frac{\varepsilon_m}{K_m} \neq \frac{\varepsilon_p}{K_p} \quad (17)$$

then the condition of continuity of the normal components of the current density and the displacement vector on both sides of the surface are inconsistent with one another, resulting in the formation of free ionic charge near the surface. The finite time needed for the formation of this free charge (of the order of τ_m) is in fact responsible for the Maxwell–Wagner dielectric dispersion approximately at $\omega_m = 1/\tau_m$.

The conductivity of the particle K_p must be modified, according to O’Konski [21], to include the contributions of both the volume and the surface conductivity K^σ in an effective particle conductivity K_{per} :

$$K_{\text{per}} = K_p + \frac{2K^\sigma}{a} \quad (18)$$

where K^σ includes both the surface conductivity within the stagnant layer and the surface conductivity of the diffuse layer beyond the plane of shear.

Furthermore, both conductivity and dielectric constant can be considered as parts of the complex dielectric permittivity of any of the system components, as shown by Eq. (12) for the dispersion medium, and by

$$\varepsilon_{\text{per}}^*(\omega) = \varepsilon_p - \frac{iK_{\text{per}}}{\omega} \quad (19)$$

for the particle. In terms of these quantities, the Maxwell–Wagner–O’Konski theory gives the following expression for the complex dipole coefficient of a disperse particle:

$$\gamma_p^*(\omega) = a^3 \frac{\epsilon_{\text{pef}}^* - \epsilon_m^*}{\epsilon_{\text{pef}}^* + 2\epsilon_m^*}. \quad (20)$$

In what follows, we shall consider the usual situation that the particle is non-conducting ($K_p = 0$), so that

$$K_{\text{pef}} = \frac{2K^\sigma}{a}. \quad (21)$$

Under this condition (non conducting particles with surface conductivity), the Maxwell–Wagner–O’Konski dipole coefficient tends to a frequency independent value in the low-frequency range characterized by inequality (4):

$$\gamma_p^*(\omega) = \frac{2Du - 1}{2Du + 2} \quad (22)$$

where Du is Dukhin’s number defined as:

$$Du = \frac{K^\sigma}{K_m a}. \quad (23)$$

2.3. Low-frequency dielectric dispersion caused by the concentration polarization of a thin double layer

Whereas the Maxwell–Wagner dielectric dispersion arises due to the different space dependencies of the local conductivity and permittivity, the difference in the space dependence of the relative contributions of cations and anions to the local conductivity (space dependence of the *transfer numbers* of cations and anions) gives raise to variations in the local electrolyte concentration. This in turn gives rise to the *volume diffusion mechanism of the low-frequency dielectric dispersion*. The contributions of cations $K^{\sigma+}$ and anions $K^{\sigma-}$ to the surface conductivity, are usually quite different (the contribution of counterions predominates). On the contrary, the contributions of cations K_m^+ and anions K_m^- to the bulk electrolyte solution conductivity K_m have comparable values. This means that almost always, the following inequality holds:

$$\frac{K^{\sigma+}}{K_m^+} \neq \frac{K^{\sigma-}}{K_m^-}. \quad (24)$$

Under this condition, the balance between the flows of the ions of both signs is only possible at

the expense of a field-induced variation of electrolyte concentration $\delta c(\vec{r})$ near the polarized particle. The ionic diffusion caused by $\delta c(r)$ gives raise to a low-frequency dependence of the particle’s dipole coefficient, $\gamma_p^*(\omega)$.

For a z -valent symmetrical electrolyte, the values of $K_m^\pm$ are determined by the expressions:

$$K_m^\pm = \frac{z^2 e^2 n_0}{kT} D^\pm \quad (25)$$

where n_0 is the anion or cation number concentration in the bulk electrolyte solution. Eq. (25) correspond to the well-known expression for the electrolyte solution conductivity:

$$K_m = K_m^+ + K_m^- = \frac{z^2 e^2 n_0}{kT} (D^+ + D^-). \quad (26)$$

As for the expressions for the contributions of the two types of ions to the surface conductivity $K^{\sigma\pm}$, they depend on the model used for the description of the kinetic properties of the double layer. In the framework of the *standard model*, they are given by:

$$K_d^{\sigma\pm} = K_m^\pm \left\{ 2 \frac{\exp\left(\mp \frac{z\tilde{\zeta}}{2}\right) - 1}{\kappa} + 3m^\pm \left[2 \frac{\exp\left(\mp \frac{z\tilde{\zeta}}{2}\right) - 1}{\kappa} \pm \frac{z\tilde{\zeta}}{\kappa} \right] \right\}. \quad (27)$$

$$\tilde{\zeta} = \frac{e}{kT} \zeta \quad (28)$$

$$m^\pm = \frac{2\epsilon_m}{3\eta D^\pm} \left(\frac{kT}{ze} \right)^2 \quad (29)$$

which correspond to Bikerman’s formula to the surface conductivity [27]:

$$K_d^\sigma = \frac{2K_m}{\kappa} \left\{ \frac{D^+}{D^+ + D^-} \left[\exp\left(-\frac{z\tilde{\zeta}}{2}\right) - 1 \right] (1 + 3m^+) + \frac{D^-}{D^+ + D^-} \left[\exp\left(+\frac{z\tilde{\zeta}}{2}\right) - 1 \right] (1 + 3m^-) \right\}. \quad (30)$$

In Eqs. (27) and (30) the lower index d denotes that the surface conductivity is solely due to ions in the *diffuse* part of the double layer. Nevertheless, in the framework of the so-called *generalized standard model*, the surface conductivity expression includes an additional term that accounts for

the existence of conduction behind the plane of shear.

The analytical theory of the low-frequency dipole coefficient in AC fields was elaborated in [1,3]. It is based, first of all, on the local equilibrium approximation among the field-induced variations of the parameters of the polarized double layer and the field-induced variations of the electrolyte solution concentration adjoining the double layer. This approximation is well justified when:

(i) Due to the small thickness of the double layer,

$$\kappa a \gg 1 \quad (31)$$

the diffuse layer near every small portion of the particle's surface may be considered as a small, and hence a quasi-equilibrium part of a large system, represented by the adjoining solution.

(ii) Due to the low frequency of the external field, i.e. when the inequality (4) is fulfilled, the period of the field greatly exceeds the time needed for the re-establishment of the screening charge of the diffuse layer.

The solution of the standard electrokinetic equation system obtained in [1–3] using the local equilibrium approximation, can be expressed in the following compact form for the low-frequency dipole coefficient [6,26]:

$$\begin{aligned} \gamma_p^*(\omega) &= \frac{2Du(\zeta) - 1}{2Du(\zeta) + 2} - \frac{3(R^+ - R^-)H}{2B} \\ &+ \frac{3(R^+ - R^-)H}{2B} \frac{i\omega\tau_\alpha}{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha} + i\omega\tau_\alpha} \end{aligned} \quad (32)$$

where

$$Du(\zeta) = \frac{K_d^\sigma}{K_m a} \quad (33)$$

$$R^\pm = \frac{2K_d^{\sigma\pm}}{K_m^\pm a} \quad (34)$$

$$\tau_\alpha = \frac{a^2(D^+ + D^-)S}{4D^+ D^-} \quad (35)$$

$$A = 4Du(\zeta) + 4 \quad (36)$$

$$\begin{aligned} B &= (R^+ + 2)(R^- + 2) - U^+ - U^- \\ &- (U^+ R^- + U^- R^+)/2 \end{aligned} \quad (37)$$

$$S = \frac{A}{B} \quad (38)$$

$$H = \frac{(R^+ - R^-)(1 - z^2 \Delta^2) - U^+ + U^- + z\Delta(U^+ + U^-)}{A} \quad (39)$$

$$U^\pm = \frac{48m^\pm}{\kappa a} \ln \left[\cosh \frac{z\zeta}{4} \right] \quad (40)$$

$$\Delta = \frac{D^- - D^+}{z(D^- + D^+)} \quad (41)$$

while η is the viscosity of the suspending medium, k is the Boltzmann constant, and T the absolute temperature.

The first addend in the right-hand side of Eq. (32) coincides with Eq. (22) and represents simultaneously the high-frequency limit of the low-frequency dispersion of the dipole coefficient, and the low-frequency limit of the Maxwell–Wagner–O’Konski dispersion. The second addend represents the amplitude of the variation of the dipole coefficient corresponding to the low-frequency dispersion. Hence, the first two addends taken together, represent the low-frequency limit of the dipole coefficient with respect to both the low-frequency and the Maxwell–Wagner–O’Konski dispersions. The third frequency-dependent addend, provides the frequency dependent transition of the dipole coefficient between its low- and high-frequency limits.

The quantity S weakly depends on the Dukhin number and the product κa , being always of the order of unity (except for very small κa):

$$S \cong 1. \quad (42)$$

Hence, a comparison of Eqs. (3) and (35) leads to the important conclusion that:

$$\frac{\tau_\alpha}{\tau_m} \approx (\kappa a)^2. \quad (43)$$

This means that for thin double layers ($\kappa a \gg 1$), the characteristic frequency for the low-frequency dispersion is $(\kappa a)^2$ times lower than that of the Maxwell–Wagner dispersion.

2.4. Inclusion of the surface conductivity of the stagnant layer

Eqs. (30), (32–40) represents the characteristics of the low-frequency dispersion of the dipole coefficient calculated in the frame of the standard electrokinetic model. According to the latter, the only parameter of the solid/liquid interface that is needed to account for LFDD is the ζ -potential. However, with the exception of rare cases (one example is presented in [26,28] for the suspension of AgI), the contribution K_d^σ of the diffuse layer (expressed by Bikerman's formula), is only a part, and in many cases even a small part, of the total value of the surface conductivity. This means that in the majority of real systems, the main part of the total surface conductivity is due to counterions in the inner part of double layer, which are responsible for the conductivity behind the slipping plane. The latter includes the contribution K_i^σ of mobile counterions, such as ions localized in the hydrodynamically immobile (due to surface roughness or polymer adsorption) part of the diffuse layer, and to mobile counterions in the Stern layer.

In order to deduce an expression for the low-frequency behavior of the dipole coefficient taking into account the surface conductivity of the stagnant layer, let us examine the physical meaning of the different terms intervening in Eq. (32). The most important are the parameters $R^\pm$, which are related to the surface and electrolyte solution conductivities of cations and anions [Eqs. (34) and (30)]. The parameters $U^\pm$ [Eq. (36)] which characterize the contribution of the convection induced by $\delta c(\vec{r})$ (capillary–osmotic convection) to the partial surface currents of cations and anions, have a very little influence on the overall values of these currents.

Neglecting the tangential flow of coions in the double layer as compared to the flow of counterions, as well as the capillary osmotic term for counterions, all the magnitudes entering in the expression of the dipolar coefficient can be expressed in terms of the surface conductivity and the electrolyte solution properties:

$$(R^+ - R^-)H \Rightarrow \frac{Du^2(\zeta)}{Du(\zeta) + 1} \frac{(1 - z^2\Delta^2)}{t_c^2} \quad (44)$$

$$A = 4Du(\zeta) + 4 \quad (45)$$

$$B \Rightarrow 4 \left(\frac{Du(\zeta)}{t_c} + 1 \right) \quad (46)$$

$$S \Rightarrow \frac{Du(\zeta) + 1}{\frac{Du(\zeta)}{t_c} + 1} \quad (47)$$

where

$$t_c = \frac{D_c}{D^+ + D^-} \quad (48)$$

and D_c is the diffusion coefficient of counterions.

The contribution of the surface conductivity due to ions situated behind the slipping plane, K_i^σ , can be characterized by the parameter:

$$\Theta = \frac{K_i^\sigma}{K_d^\sigma} \quad (49)$$

To emphasize the influence of K_i^σ on the Dukhin number, both arguments, ζ (for K_d^σ) and Θ (for K_i^σ) must be used:

$$Du(\zeta, \Theta) = \frac{K^\sigma}{K_{ma}} = \frac{K_d^\sigma}{K_{ma}} (1 + \Theta). \quad (50)$$

The problem of the generalization the theory of the volume diffusion mechanism by taking into account the surface conductivity due to ions situated behind the slipping plane has been dealt with in Refs. [29–31] for the case when $\kappa a \gg 1$. It can be shown that simplified formulae obtained by neglecting the coion contribution to the surface current in the double layer, and also by neglecting the contributions of the currents of capillary–osmotic origin, allow to write the low-frequency dispersion in terms of only one surface characteristic, namely $Du(\zeta, \Theta)$. The following substitutions must be carried out in Eqs. (32), (36)–(38):

$$(R^+ - R^-)H \Rightarrow \frac{Du^2(\zeta, \Theta)}{Du(\zeta, \Theta) + 1} \frac{(1 - z^2\Delta^2)}{t_c^2} \quad (51)$$

$$A = 4Du(\zeta, \Theta) + 4 \quad (52)$$

$$B \Rightarrow 4 \left(\frac{Du(\zeta, \Theta)}{t_c} + 1 \right) \quad (53)$$

$$S \Rightarrow \frac{Du(\zeta, \Theta) + 1}{\frac{Du(\zeta, \Theta)}{t_c} + 1}. \quad (54)$$

2.5. Wide-frequency range polarizability and dielectric dispersion

A general formula for the induced dipole moment will now be described that yields simultaneously the frequency dependence of both dispersion ranges, namely, the α -dispersion and the Maxwell–Wagner–O’Konski dispersion. The treatment was recently elaborated in [32], for conditions corresponding to the quasi-equilibrium polarization of the Stern layer in iso-electric point (in the absence of equilibrium diffuse layer), but it has not been applied yet in the frame of an analytical theory of thin double layer polarization. However, the coincidence [see Eq. (32)] of the high-frequency limit of the dipole coefficient obtained from the analytical theory of the low-frequency-dispersion, with the low-frequency limit of dipole coefficient obtained by using the theory of Maxwell–Wagner–O’Konski [22], creates the necessary prerequisites for a simple procedure of derivation of the general formula for the dipole coefficient γ_p^* .

In order to derive this formula, a superposition approximation will be used, whereby we simply replace the frequency-independent term $[2Du(\zeta) - 1]/[2Du(\zeta) + 2]$, representing the high-frequency limit of Eq. (32) by the Maxwell–Wagner–O’Konski dipole coefficient, $[\varepsilon_{\text{per}}^* - \varepsilon_m^*]/[\varepsilon_{\text{per}}^* + 2\varepsilon_m^*]$. After some algebra, this leads to:

$$\gamma_p^* = \frac{\frac{\varepsilon_p}{\varepsilon_m} - 1}{\frac{\varepsilon_p}{\varepsilon_m} + 2} - \frac{3}{2} \frac{\frac{\varepsilon_p}{\varepsilon_m} - 2Du}{\left(\frac{\varepsilon_p}{\varepsilon_m} + 2\right)(Du + 1)} \frac{1}{1 + i\omega\tau_{\text{MW}}} - \frac{3(R^+ - R^-)H}{2B} \frac{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha}}{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha} + i\omega\tau_\alpha} \quad (55)$$

where

$$\tau_{\text{MW}} = \frac{\varepsilon_m}{2K_m} \frac{\varepsilon_p + 2}{Du + 1} \quad (56)$$

$R^\pm$, H , S , A , and Du are given either by the standard electrokinetic model expressions (23), (32)–(41), or by Eqs. (49)–(54) if surface conductivity behind the slipping plane is taken into account.

The asymptotic validity of the superposition approximation is determined by the strong inequality expressing the very small value of relaxation time τ_{MW} (corresponding to the Maxwell–Wagner–O’Konski dispersion) as compared to τ_α , corresponding to the low-frequency dispersion (related to the volume diffusion mechanism):

$$\tau_{\text{MW}} \ll \tau_\alpha. \quad (57)$$

The superposition approximation does not take into account the mutual influence of the volume diffusion mechanism (inherent to the low-frequency dispersion) and the displacement current (inherent to the Maxwell–Wagner–O’Konski dispersion) in the process of particle polarization. But if condition (57) holds, the frequencies at which the volume diffusion mechanism is essential for the particle’s polarization ($\omega \approx 1/\tau_\alpha$) happen to be very small ($\omega \ll 1/\tau_{\text{MW}}$), so that displacement currents are negligible with respect to conduction currents. On the other hand, the frequencies at which these two types of currents are comparable ($\omega \approx 1/\tau_{\text{MW}}$) are so large that the volume diffusion mechanism has a negligible influence on the fields and currents ($\omega \gg 1/\tau_\alpha$). The value of τ_{MW} is close to the value of electrolyte relaxation time (1) for the majority of typical systems:

$$\tau_{\text{MW}} \cong \frac{\varepsilon_m}{K_m} = \tau_m. \quad (58)$$

From Eqs. (31) and (58) it follows that:

$$\frac{\tau_\alpha}{\tau_{\text{MW}}} \cong (\kappa a)^2 \gg 1. \quad (59)$$

This means that the strong inequality (57) that allows to use the superposition approximation, is closely related to the inequality (31), describing the small thickness of the Debye atmosphere as compared to particle radius. Therefore, the superposition approximation that leads to the general formula (55) for the dipole coefficient, happens to

be applicable just in the case of thin double layers, for which both theories: the theory of volume diffusion mechanism of the low-frequency dispersion and the theory of the Maxwell–Wagner–O’Konski dispersion, are applicable.

In order to check the accuracy of the wide-frequency range formula (55), the frequency depen-

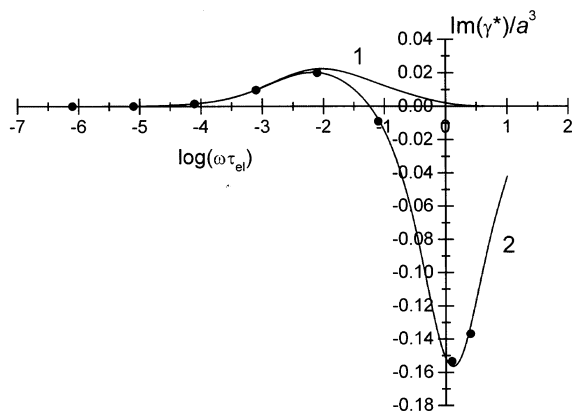


Fig. 1. Comparison between dipole coefficient data obtained by means of Eq. (55) (curve 2) with numerical calculations [25] (symbols), for a KCl solution with $\kappa a = 25.12$. Dipole coefficient data obtained with the LFDD theory [3,6] are also included (curve 1). Parameters used: $\epsilon_m = 80\epsilon_0$; $\epsilon_p = 2\epsilon_0$; $K_{\text{per}} = 0.875 \cdot 10^{-1} \text{ S m}^{-1}$; $D^+ = D^- = 2 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$; $\eta = 10^{-3} \text{ g (m s}^{-1})$; $\zeta = 100 \text{ mV}$.

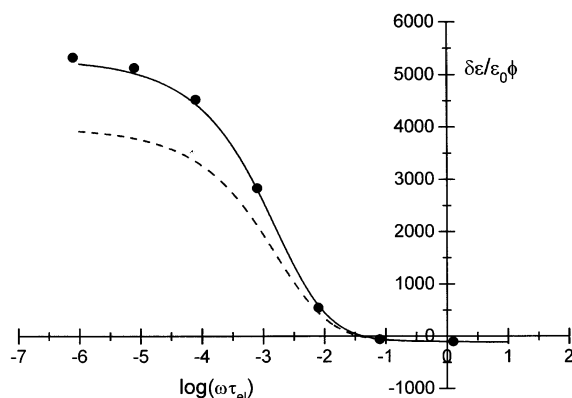


Fig. 2. Comparison of analytical (curves) and numerical (symbols) results of the wide-frequency range dependence of the suspension permittivity. Solid line: full calculation based on the standard model; dashed line: results obtained neglecting coion and capillary–osmotic contributions. Same parameters as in Fig. 1.

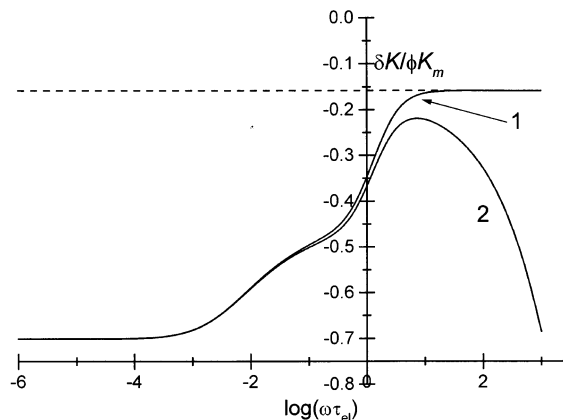


Fig. 3. Frequency dependence of the suspension conductivity. Curve 1: modified formula (61); curve 2: direct substitution of Eq. (55) into (11). The dashed horizontal line shows the high frequency limit of the Maxwell–Wagner–O’Konski dispersion of the conductivity, Eq. (62). Same parameters as in Fig. 1.

dence of the dipole coefficient, for the case of the standard model, has been compared with numerical calculations [25]. The results for the imaginary part of the dipole coefficient, $\text{Im} [\gamma_p^*(\omega)]$ are represented in Fig. 1 (curve 2). As can be seen, the analytical results are confirmed by numerical calculations, with a relative error lesser than 3% for $\kappa a > 25$, in the frequency range including both the volume diffusion and the Maxwell–Wagner dispersions. Curve 1 shows the dipole coefficient calculated using the low-frequency expression [Eq. (32)].

The frequency dependence of the suspension permittivity in the wide-frequency range that including both dispersions, can be obtained by direct substitution of Eq. (55) into Eq. (10). Fig. 2 demonstrates a good agreement of $\delta\epsilon(\omega)$ calculated in this way using the coefficients corresponding to the standard model, Eqs. (30), (32)–(41), with the numerical results of DeLacey and White [25]. The dashed curve represents the analytical results obtained using the coefficients (44)–(48) (in which the coion and the capillary–osmotic contributions to the surface ionic flows are neglected) with $\Theta = 0$.

As for the calculation of the frequency dependence of the suspension conductivity, a direct substitution of (55) into (11) leads to a wrong

behavior of $\delta K(\omega)$ for very high frequencies, exceeding ω_m (see Fig. 3, curve 2). The origin of this incorrect behavior is the high frequency asymptotic form of the last term in the right-hand side of Eq. (55), which is of $O(1/\sqrt{\omega\tau_\alpha})$ in its imaginary part. The product of this asymptotic dependence and the imaginary part of K_m^* [equal to $\omega\epsilon_m$, Eq. (11)] causes the divergence of $\delta K(\omega)$. This slowly decreasing high-frequency asymptotic term is a consequence of the approximation of local equilibrium in the theory of the volume diffusion mechanism. This approximation works well when the period of the alternating field greatly exceeds the relaxation time of the screening charge (diffuse layer) τ_m . However, when the frequency reaches values comparable to τ_m and higher, the diffuse layer can not follow the field-induced concentration variations. Furthermore, for these high frequencies, the concentration variations themselves occur inside the diffuse layer, and it is no longer possible to assume the existence of 'local equilibrium of the field-induced variations of the parameters of the polarized double layer with the field-induced variations of the electrolyte solution concentration adjoining the double layer'.

Due to the these two factors, the slowly decreasing high frequency asymptotic form of $\gamma_p^*(\omega)$, given in Eqs. (32) or (46), must fully vanish for high frequencies, i.e., above ω_m . The influence of the concentration polarization on the dipole coefficient, is reflected by the last addend in the right hand side of Eq. (55). Hence, a possible way to take into account these arguments in the expression for the wide frequency-range dipole coefficient $\gamma_p^*(\omega)$, is to modify Eq. (55) multiplying its last addend by some function $F(\omega\tau_m)$, which is equal to unity at low frequencies and rapidly decreases when the frequency increases. A suitable function is:

$$F(\omega\tau_{cl}) = e^{-\omega\tau_{cl}} = \exp\left(-\omega\frac{\epsilon_m}{K_m}\right). \quad (60)$$

This dependence guarantees the correct influence of the concentration polarization when condition (3) is fulfilled (i.e. when the double layer may be considered as locally in equilibrium), and simultaneously eliminates the parasitic, slowly de-

creasing high frequency behavior of $\gamma_p^*(\omega)$. The resulting modified expression for the wide-frequency range dipole coefficient is:

$$\begin{aligned} \frac{\gamma_p^*}{a^3} = & \frac{\epsilon_p}{\epsilon_m} - 1 - \frac{3}{2} \frac{\epsilon_p - 2Du}{\epsilon_m} \frac{1}{\left(\frac{\epsilon_p}{\epsilon_m} + 2\right)(Du + 1)} \frac{1}{1 + i\omega\tau_{MW}} \\ & - \frac{3(R^+ - R^-)H}{2AS} \frac{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha}}{1 + \sqrt{2/S}\sqrt{i\omega\tau_\alpha} + i\omega\tau_\alpha} \\ & \times \exp\left(-\omega\frac{\epsilon_m}{K_m}\right). \end{aligned} \quad (61)$$

In the high frequency range, where the relative difference between Eqs. (55) and (61) is significant, the absolute value of $\text{Im}[\gamma_p^*(\omega)]$ is very small, and the difference cannot be appreciated in Fig. 1. The particle's contribution to the dielectric permeability, $\delta\epsilon(\omega)$, calculated combining Eq. (10) and either Eq. (55) or (61), are also very close and indistinguishable in Fig. 2.

3. Discussion

The frequency dependence of the particle's contribution to the conductivity $\delta K(\omega)$, calculated combining Eqs. (11) and (55) for the case of the standard model, is represented by curve 2 in Fig. 3. The dashed horizontal line shows the value of the well-known (see, for example [33]) high-frequency limit of the Maxwell–Wagner–O'Konski conductivity dispersion, δK_{MW}^h :

$$\frac{\delta K_{MW}^h}{\phi K_m} = \frac{3(\epsilon_p - \epsilon_m)}{\epsilon_p + 2\epsilon_m} + \frac{9\epsilon_m \left(\frac{K_{p\text{ef}}}{K_m} \cdot \epsilon_m - \epsilon_i\right)}{(\epsilon_p + 2\epsilon_m)^2}. \quad (62)$$

For non-conducting particles with thin diffuse double layer, aDu must be substituted for the quotient $K_{p\text{ef}}/K_m$ appearing in this expression. It is clear from Fig. 3 that the modified formula (61) effectively eliminates the parasitic high-frequency behavior of $\gamma_p^*(\omega)$ and $\delta K(\omega)$, while simultaneously maintaining the correct low-frequency behavior. This is demonstrated by the fact that $\delta K(\omega)$ reaches exactly the high-frequency limit (62).

The low-frequency limit of $\delta K(\omega)$ is, in agreement with Eq. (11), proportional to the low frequency limit of $\gamma_p^*(\omega)$, which is represented by the sum of the first two terms in the right-hand side of Eq. (32). Expressing this limit in terms of Dukhin's number with the help of Eqs. (44)–(47), and using Eq. (41) (for Δ) and (48) (for t_c), leads to:

$$\begin{aligned} \delta K|_{\omega \rightarrow 0} &= 3\phi K_m \gamma_p^*(\omega)|_{\omega \rightarrow 0} \\ &= 3\phi_p \left[\frac{Du - 1}{4Du + 2} \pm \frac{3z\Delta Du^2}{(4Du + 2)(Du + t_c)} \right]. \end{aligned} \quad (63)$$

The upper (lower) sign in the right hand side of this expression, corresponds to a positively (negatively) charged particle. The first addend between the square brackets corresponds to the static (direct current) dipole coefficient. It should be emphasized that, according to (63), the low-frequency limits of $\delta K(\omega)$ and $\gamma_p^*(\omega)$ only coincide with their corresponding static values for the case $\Delta = 0$, i.e. when the diffusion coefficients of coions and counterions are the same.

The cause of this difference is the contribution to the dipole coefficient of the volume charge distribution that arises in the non-static clouds of electrolyte concentration variations δc , which are due to the different values of the anion and cation

diffusion flows. Formally, this volume charge manifests itself by the fact that the alternating field-induced potential distribution outside the double layer is described by the Poisson (not Laplace's) equation with its right-hand side proportional to $\sqrt{\omega\tau_\alpha}\delta c\Delta$ (see, for example, [1,3,6]). It is evident that since the ions with the highest mobility leave faster the regions with positive δc , the sign of the volume charge will coincide with the sign of the ion with lesser mobility (correspondingly the field-induced volume charge in the regions with negative δc will have the sign of the most mobile ion).

The field-induced concentration variation δc is positive near that particle pole where counterions move from the double layer into the bulk (and negative near the opposite pole where counterions move from the bulk towards the double layer. Therefore, the low-frequency limits of the dipole coefficient and of the suspension conductivity increment are larger than the corresponding static values if coions are more mobile than counterions, and vice versa. Consequently, in the case of more mobile coions, the low-frequency limit of the conductivity increment may substantially exceed the value $\frac{3}{4}\phi_p K_m$, which is the maximum value attainable by the static conductivity increment. The influence of the difference of diffusion coefficients on the low-frequency conductivity is shown in Fig. 4 for the case of a negatively charged particle.

Another interesting behavior corresponds to the high-frequency increment of the suspension conductivity, which may reach very high values exceeding by several decades the value $\frac{3}{4}\phi_p K_m$. This is to be expected for $\omega \gg \omega_{MW}$ if $Du(\zeta, \Theta)$ is so large that it gives rise to the inequality $\tau_{mW} \gg \tau_m$, [see Eq. (57)]. This phenomenon, shown in Fig. 5, is the result of the large real value of the product of the imaginary parts of the medium conductivity and the dipole coefficient.

As a final remark it is worth noting that while the theory presented in this work was developed for homogeneous non-conducting particles, some of the obtained results are applicable to more complex systems such as conducting particles surrounded by an insulating membrane (biological

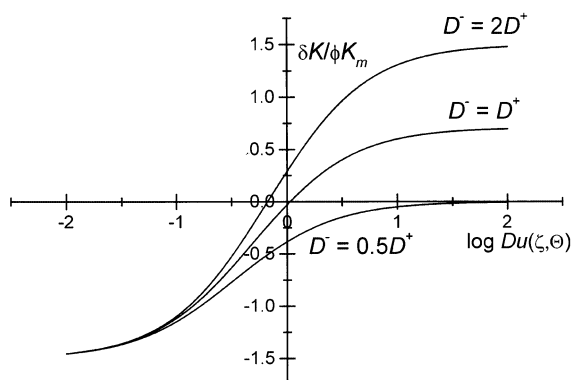


Fig. 4. Dependence of the low-frequency limit of the conductivity increment on Dukhin's number for the case of positive counterions. Same parameters as in Fig. 1 with the exception of coion diffusion coefficients, D^- , which are shown in the figure.

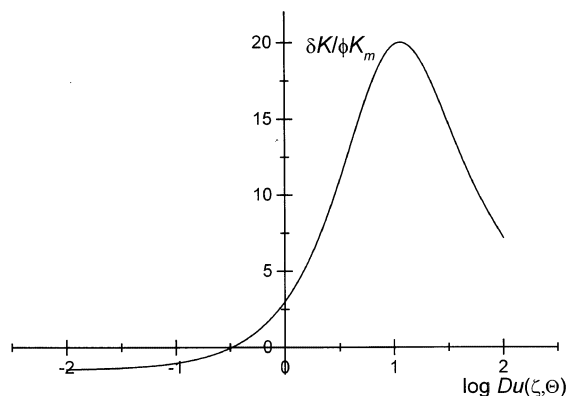


Fig. 5. Dependence of the high-frequency ($\omega = 10\omega_m$) increment of the suspension conductivity on Dukhin's number. Same parameters as in Fig. 1.

cells, for example). The limits of applicability are determined by the characteristic frequency associated to the charging of the cell membrane, which is of the order of [33,34]:

$$\omega_\beta = \frac{\frac{K_m h}{\varepsilon_{\text{mem}} a}}{\frac{K_m}{K_i} + \frac{1/2}{1 + Du(\zeta, \Theta)}} \quad (64)$$

where h is the thickness of the membrane, ε_{mem} its absolute permittivity, and K_i the conductivity of the internal medium. The results deduced for the low-frequency dispersion can be used for these systems as long $\omega < \omega_\beta$. On the contrary, the high-frequency results cannot be used in the form presented here, since at higher frequencies the capacitive current across the membrane becomes significant and the electric response of the system becomes dependent on the inner conductive properties that were not considered in this work. An approximate wide-frequency theory for these systems is presented in [34].

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